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SHOA-Adaptive: A Spintronic Framework for Programmable Self-Assembly

and Acoustic Resonance Molding of Geopolymer Nanocomposites

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Abstract

We introduce SHOA-Adaptive, a novel class of programmable

geopolymer

materials capable of controlled self-assembly under a quantum-resonant

acoustic impulse. The key innovation is the embedding of

superparamagnetic nanoparticles ("SHOA-centers") into a mineral matrix,

which function as quantum sensors and actuators obeying the Spiked

Voting protocol. An extended Hamiltonian incorporating acoustic

modulation, optimized via GRAPE, drives the material into a

geometrically optimal state, achieving micron-scale precision without

mechanical processing. This work presents a verifiable engineering

principle for quantum-resonance molding.

1. Introduction: Programmable Matter

The SHOA project (Fedotov, 2026) has evolved from self-healing

(Classic) through learning (Neuro) to non-invasive neural readout

(NeuroSHOA). SHOA-Adaptive represents the next logical step:

programmable self-assembly. We demonstrate that by introducing

quantum-active nanoparticles into a viscous mineral matrix and applying

an optimized acoustic impulse, the material can be forced to adopt a

predetermined macroscopic geometry.

2. Mathematical Model: The Extended

Hamiltonian

The total Hamiltonian of the SHOA-center ensemble embedded in a

viscoelastic matrix is:

$$H_{\text{Adaptive}} = H_{\text{spin}} + H_{\text{dip}} + H_{\text{acoustic}}(t) + H_{\text{geometry}}$$

The principal extension is $H_{\text{acoustic}}(t)$, which modulates the energy

gap of each center via a GRAPE-optimized ultrasonic waveform:

$$H_{\text{acoustic}}(t) = \sum_i \Omega_i(t) \sigma_z^i(i). \text{ The environment's dynamics are}$$

described by a self-consistent equation of motion that links the

quantum force (the gradient of $\langle H_{\text{dip}} \rangle$) with classical elastic

relaxation (the gradient of H_{geometry}).

3. Experimental Realization: Laboratory Protocol

We propose a concrete, reproducible experiment:

1. Synthesis: Superparamagnetic Fe_3O_4 nanoparticles ($\sim 10\text{-}15\text{ nm}$) are

introduced into a geopolymer mixture (metakaolin + alkaline solution).

2. Scanning: The geometry of a reference block is read by a laser

profilometer.

3. Activation: A GRAPE-optimized ultrasonic impulse is applied,

causing adaptive flow.

4. Fixation: The mixture solidifies, and the final shape is compared

to the

reference.

4. Technological Potential and Illustration

SHOA-Adaptive geopolymers enable the formation of objects with micron

precision without mechanical processing, opening a new era in

construction, restoration, and complex architectural design. If such a

mechanism were known in antiquity, it could naturally explain the enigma

of polygonal masonry, which has challenged academic science for

decades. Our method does not claim this, but serves as a striking

illustration of its potential.

5. Conclusion

We have presented a new engineering principle: the controlled quantum-resonance self-assembly of minerals. SHOA-Adaptive is a technology ready for laboratory verification.